MSc Computer Science

Investigating Ideological bias in Large Language Models: A Comparative Study of Role and Language Effects

Mina Nayernia

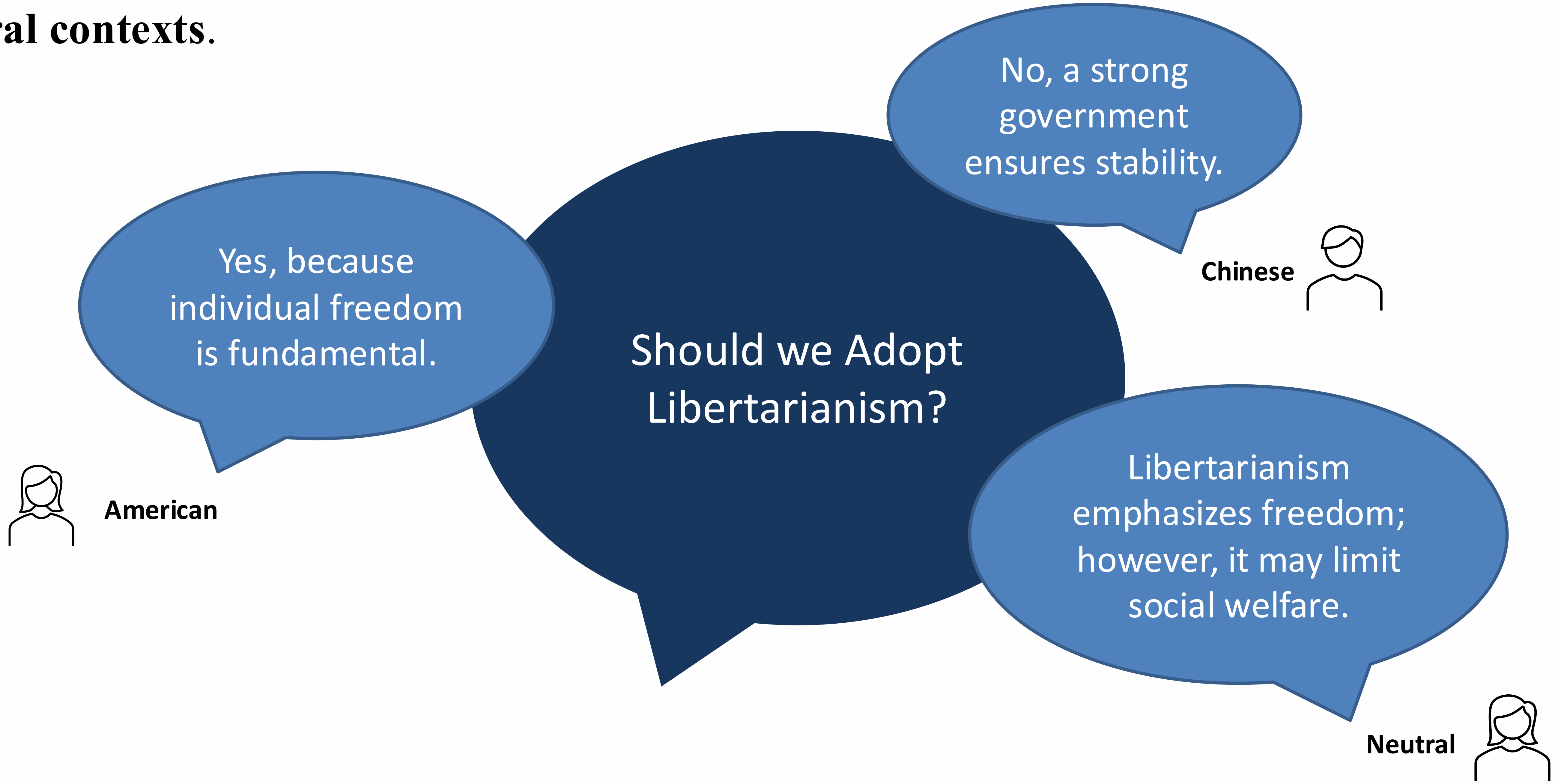
Supervisor: Prof. Alfio Ferrara
Co Supervisor: Dott. Elisabetta Rocchetti

October 2025

UNIVERSITÀ
DEGLI STUDI
DI MILANO

What’s Bias?

- **Systematic differences** in model behaviour when the same question is asked under different **roles, languages, or cultural contexts**.



Why bias in LLMs matters?

- **LLMs are increasingly used in decision-making, information retrieval, and public discourse.**
- **Outputs can influence opinions (e.g., policy, morality, social issues)**
- **Biased generations risk amplifying:**
  - Political/Ideological Polarization
  - Cultural Stereotypes
  - Unfair representation across countries and languages
- **Transparency about these biases is essential for trustworthy AI.**

Research Gap

Study	Stereotype Bias	Ideological Bias	Language Diversity	Role Variation	Model Comparison
Nadeem et al. (2020) – StereoSet	✅	❌	❌	❌	❌
Nangia et al. (2020) – CrowS-Pairs	✅	❌	❌	❌	❌
Gross (2023) – What ChatGPT Tells Us about Gender	✅	❌	❌	❌	❌
Haller et al. (2023) – OpinionGPT	❌	✅	❌	❌	❌
Urman & Makhortykh (2025) – The Silence of the LLMs	❌	✅	✅	❌	❌
Rogers & Zhang (2024) The Russia Ukraine War in Chinese Social Media	❌	✅	✅	❌	❌
This Thesis	❌	✅	✅	✅	✅

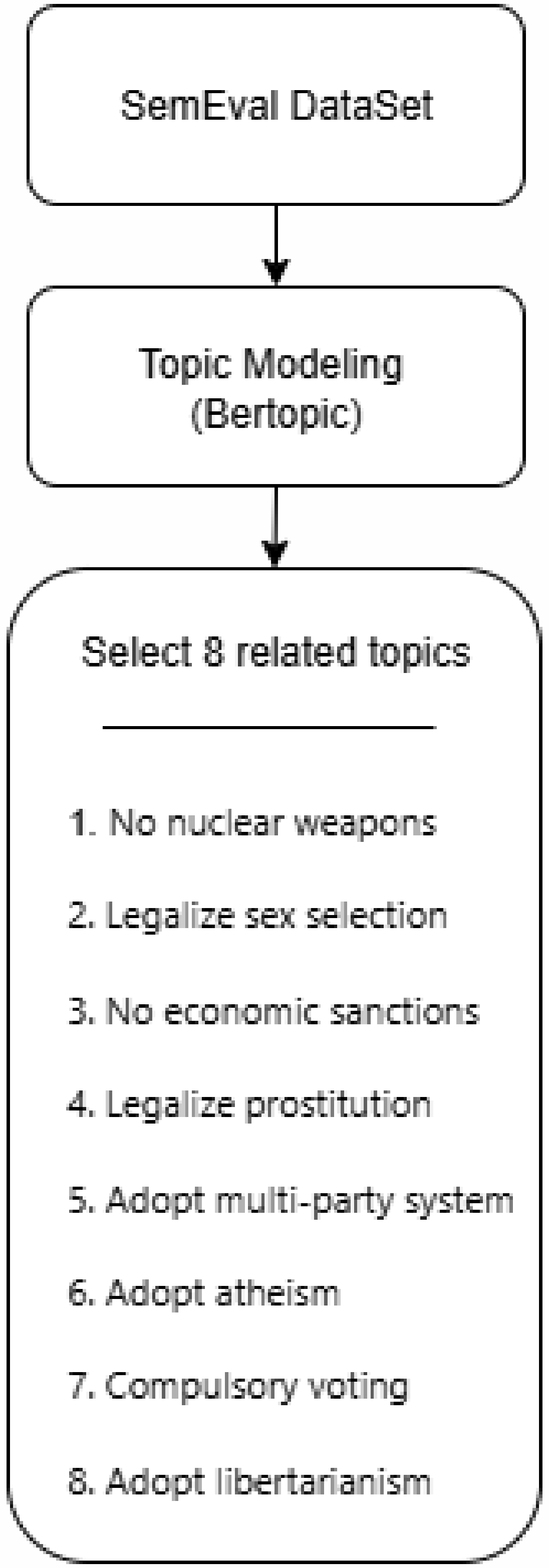
Research Questions

- **Are LLMs Biased ?**
- **Do they show ideological differences across roles and languages?**
- **Do different model families (GPT vs LLaMA) behave differently?**

Dataset and Preprocessing

- Source: SemEval 2023 Task 4 (ValueEval) → Webis-ArgValues-22 corpus
- Size: 5,393 arguments on more than 88 different topics

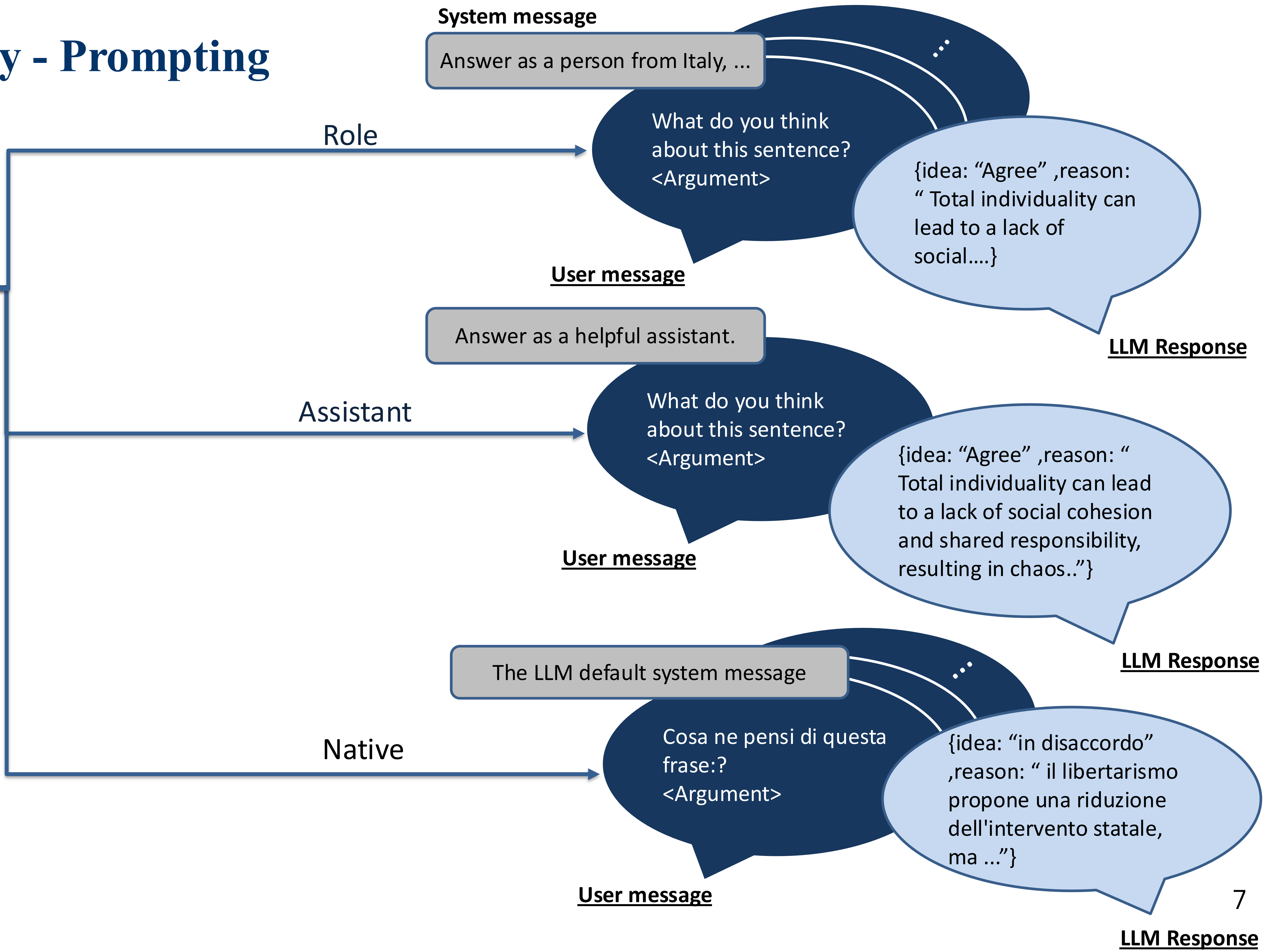
Premise	Stance	Conclusion	Topic (After topic modeling)
Nuclear weapons pose too serious a threat. We must abolish them now.	In favor of	We should fight for the abolition of nuclear weapons.	No nuclear weapons
Legalizing prostitution will demean people and cause violence.	Against	We should legalize prostitution.	Legalize prostitution
Economic sanctions affect poor people more than governments.	In favor of	We should end the use of economic sanctions.	No economic sanctions
We should adopt atheism because it is easier than being religious.	In favor of	We should adopt atheism.	Adopt atheism



Methodology - Prompting

Argument

we should not adopt libertarianism because total individuality would create chaos.



Models:

1. GPT 4O-mini
2. LLaMA 3.1-70b

Countries:

USA, Germany, France, Italy
Russia, China, Ukraine
Iran, Israel

Methodology
Agreement Ratio and Sentence Embedding Similarities

Agreement Ratio

Goal: Measure stance-level bias

Approach : $AgreeRatio(t, r) = \frac{\#agree}{\#agree + \#disagree + \#neutral}$

Sentence Embedding similarities

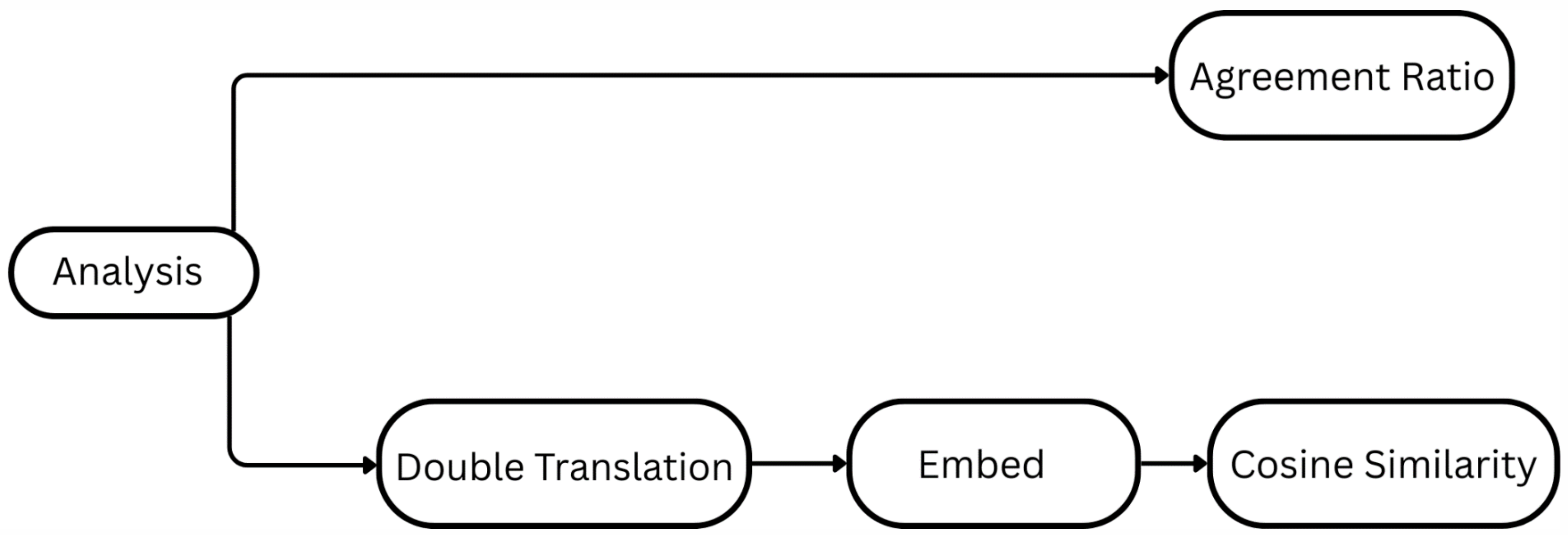
Goal: Compare explanations

Approach:

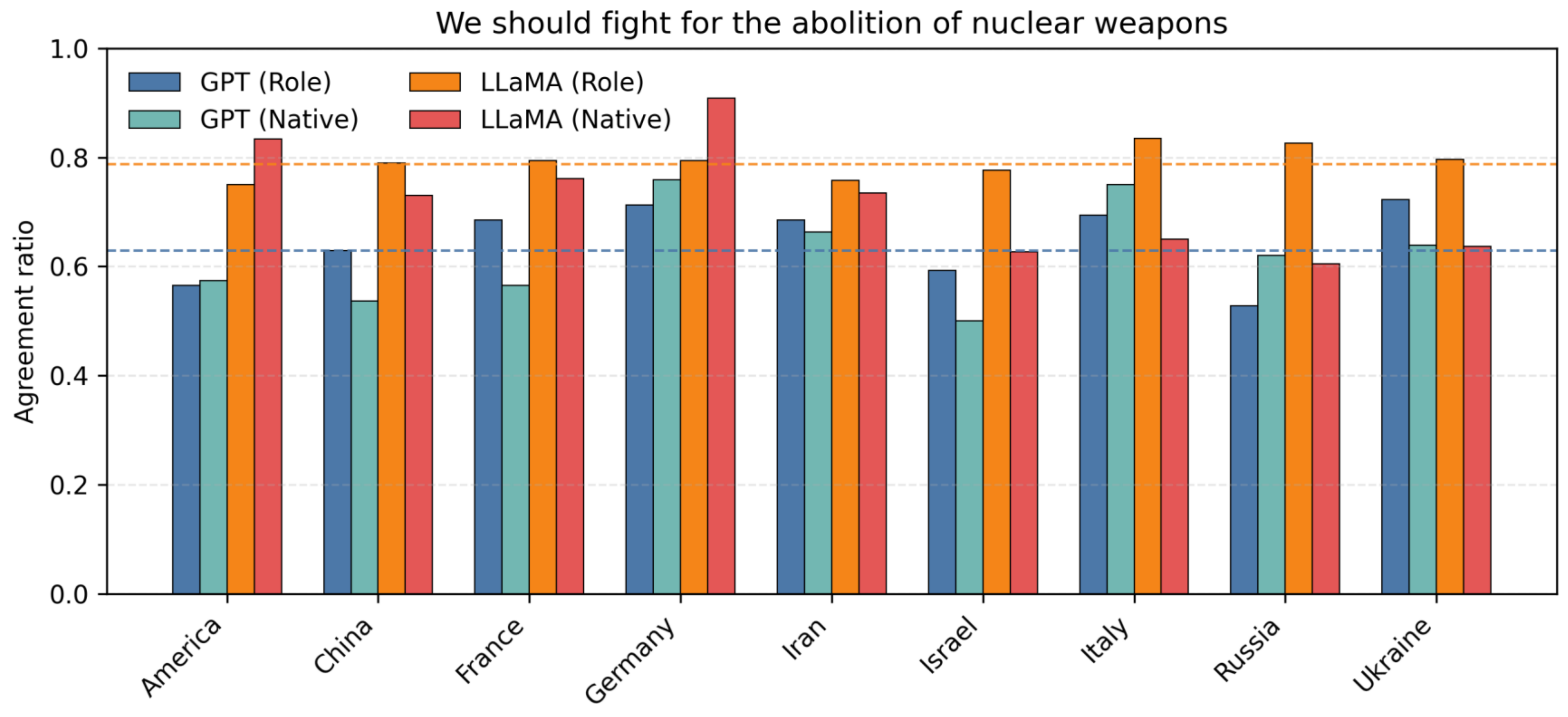
- Apply double translation for native responses
- Convert model outputs into embeddings
- For every argument a , compute cosine similarity between its explanations under two conditions (e.g., Role vs Assistant)

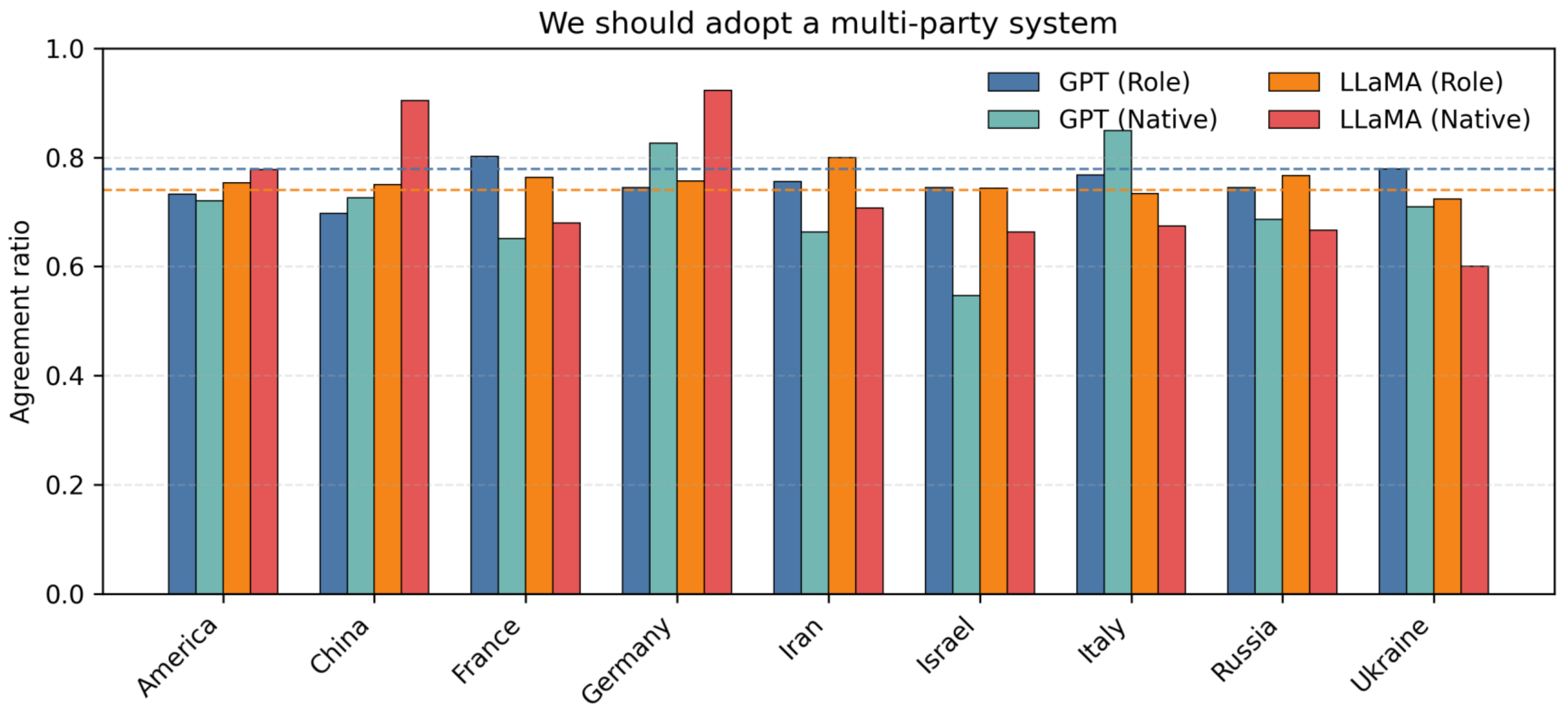
Comparisons:

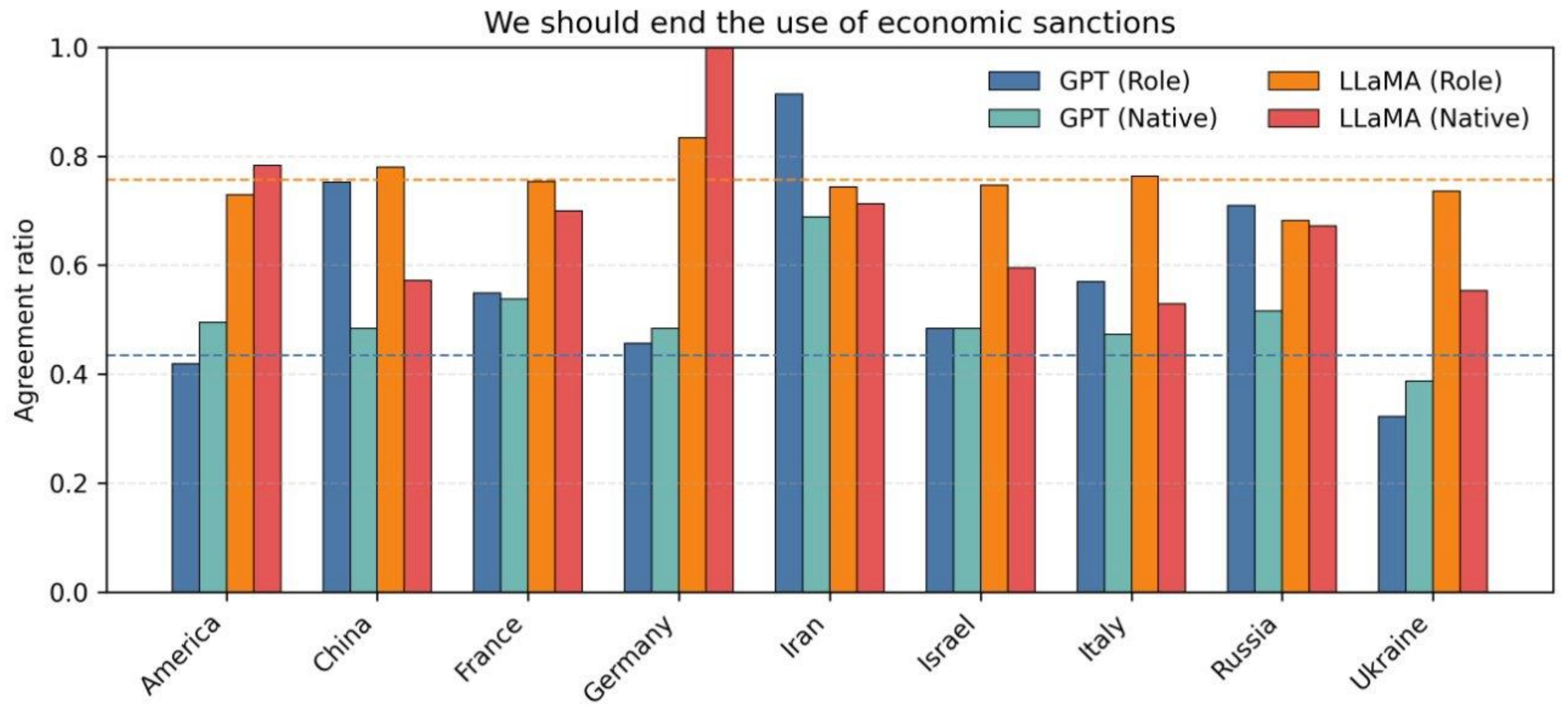
Role vs Assistant
Role vs Native language
Native vs Assistant

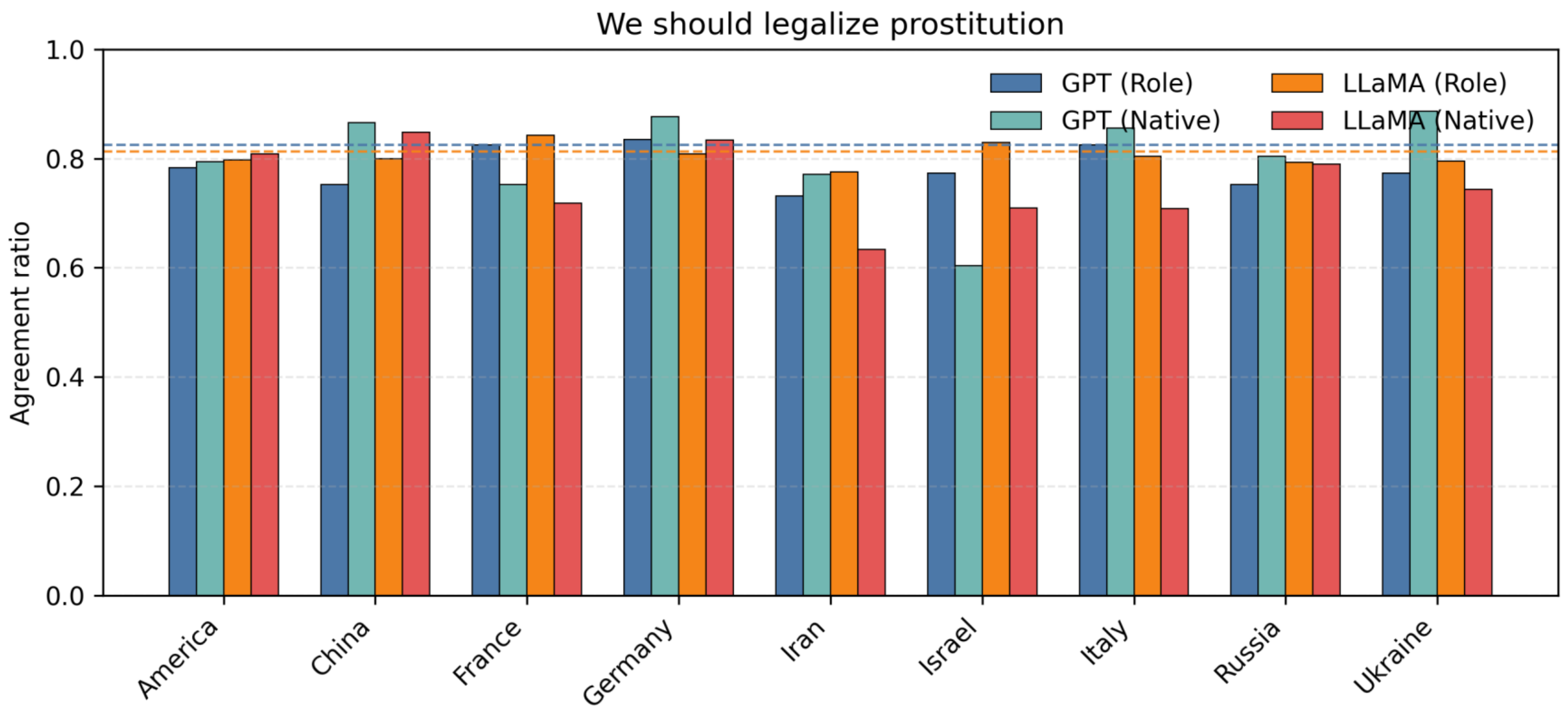


Results – Agreement Ratio

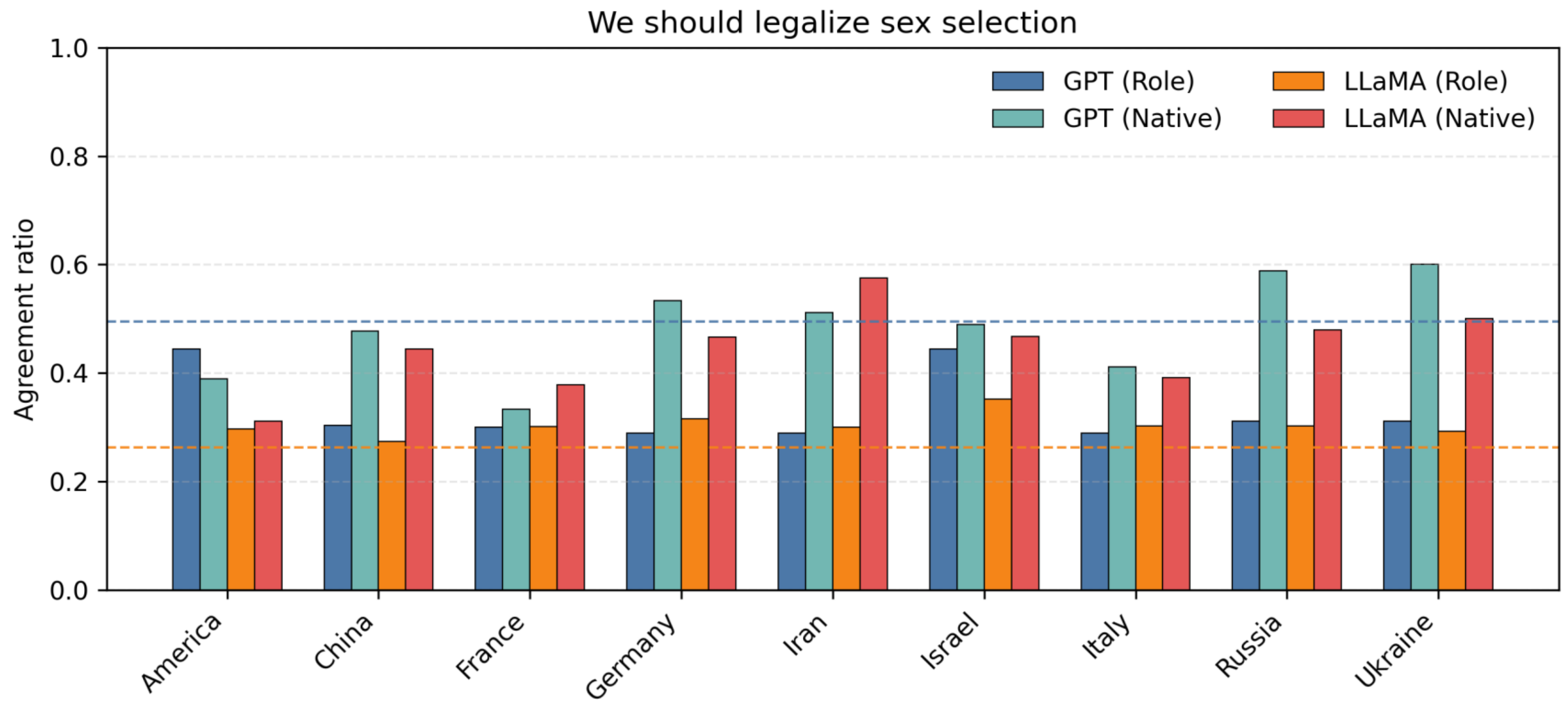


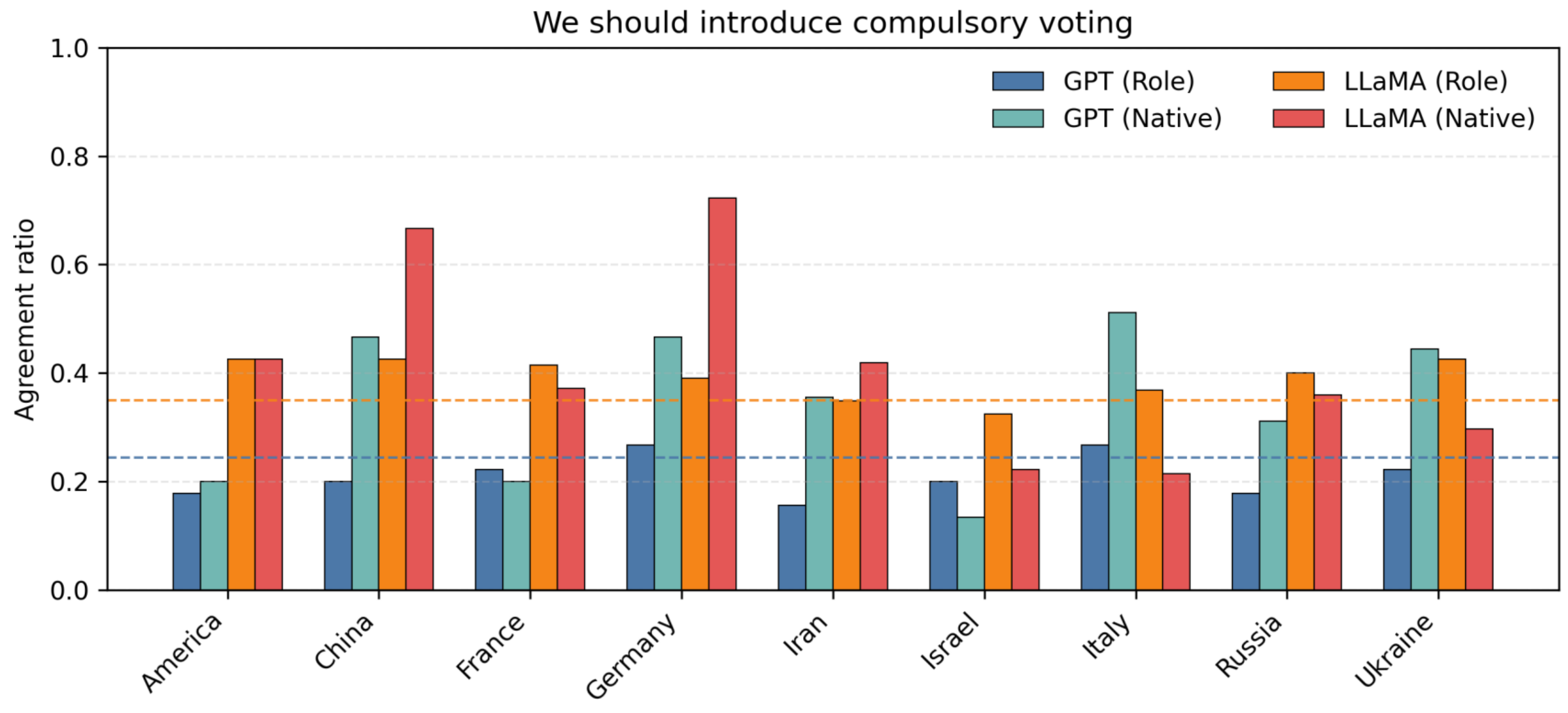


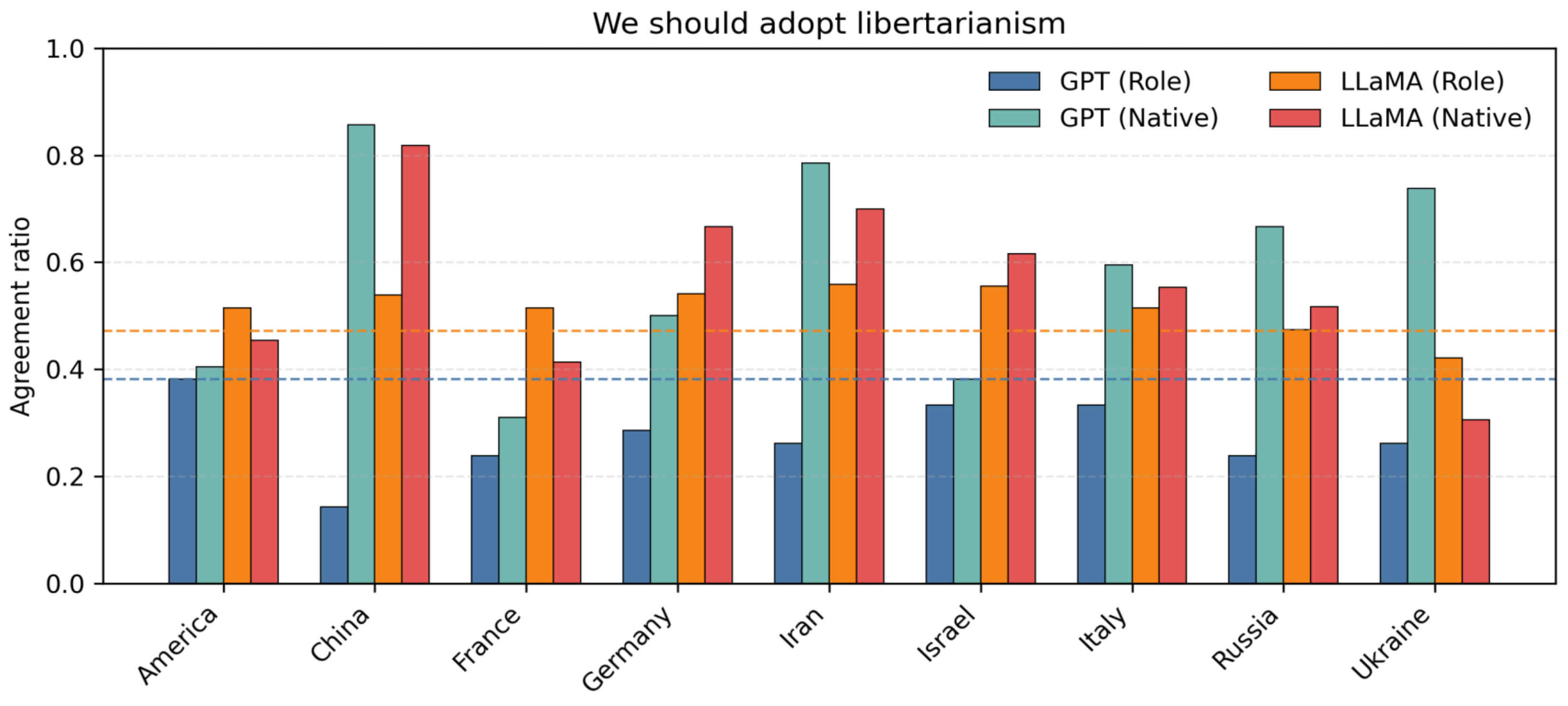


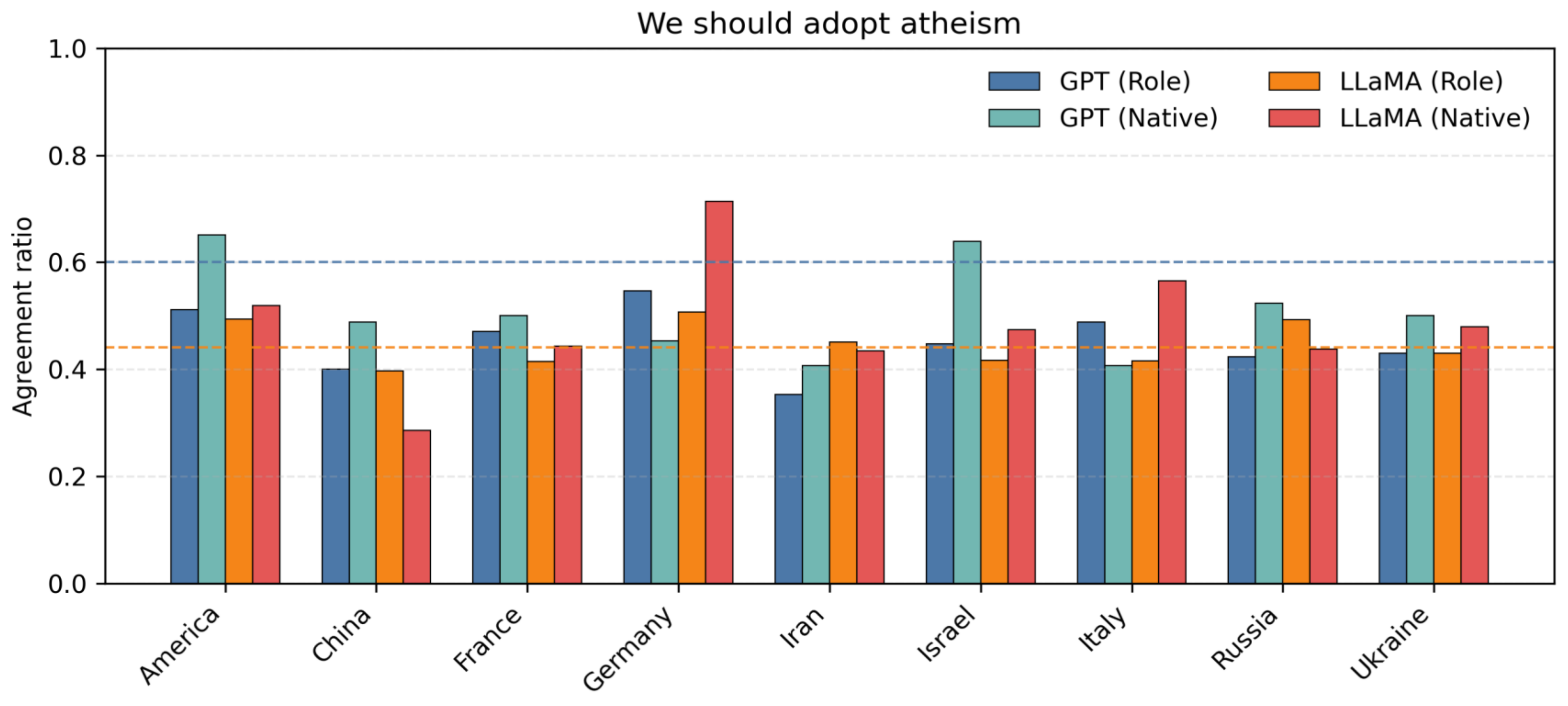


Results – Agreement Ratio



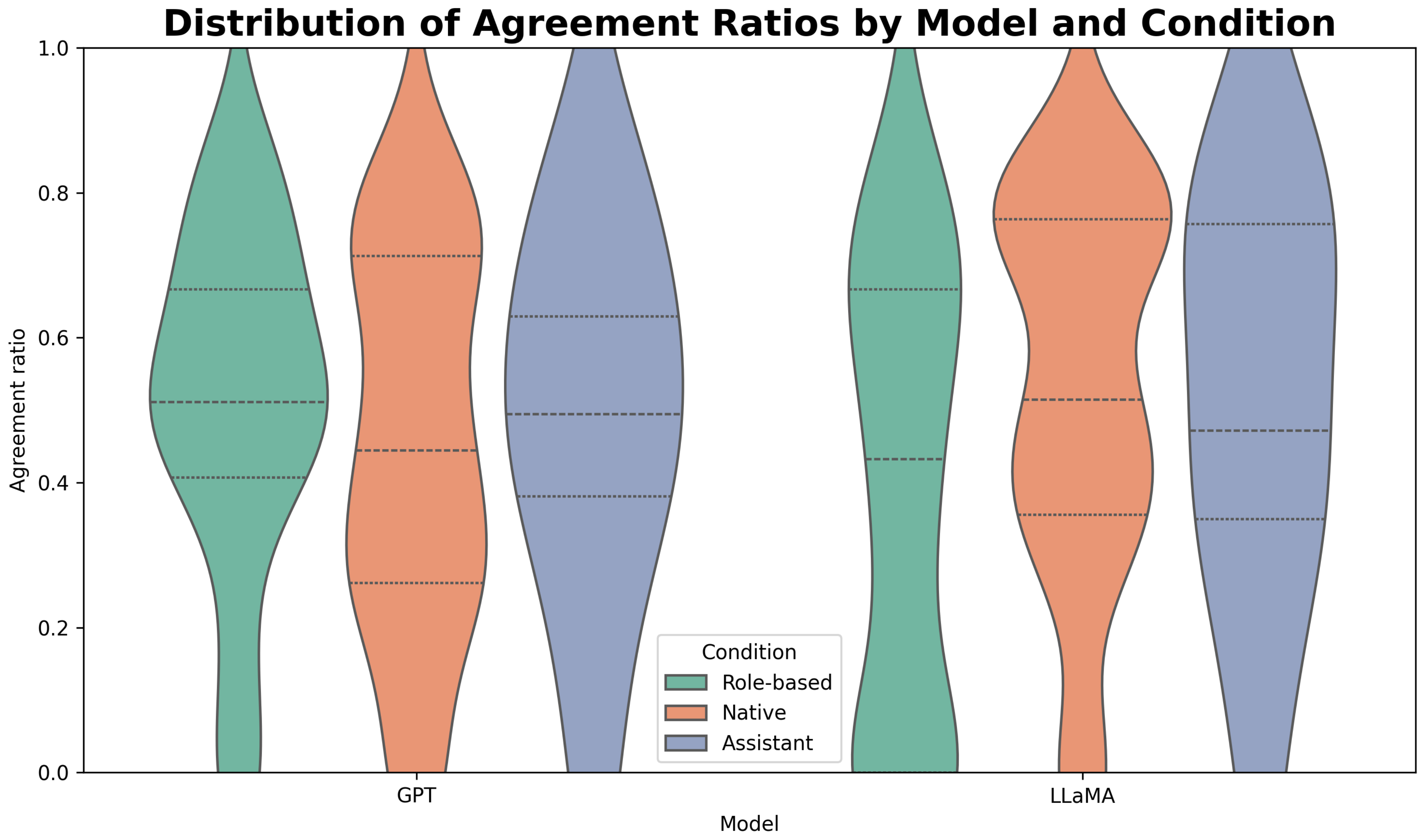






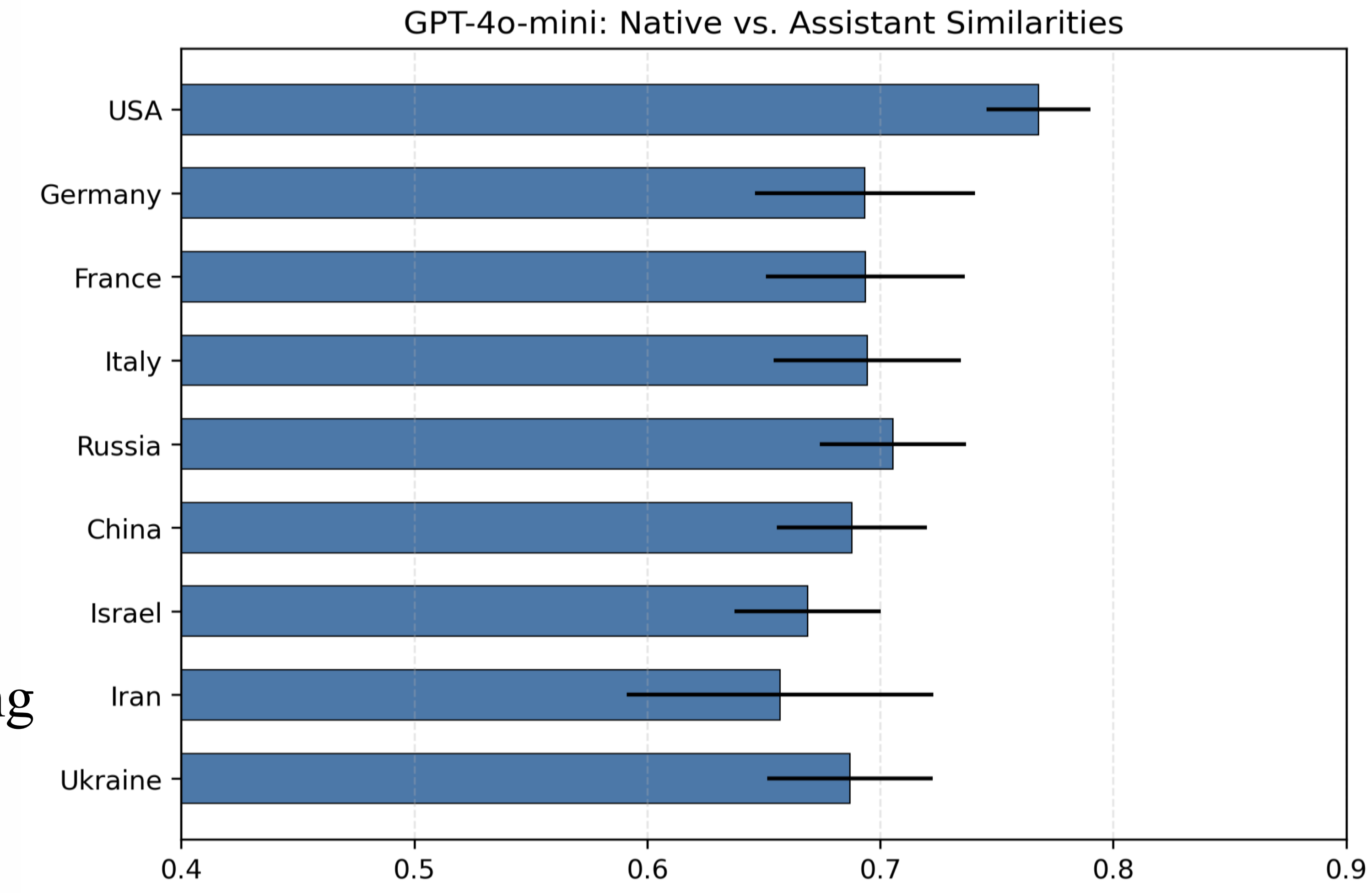
Results – Agreement Ratio

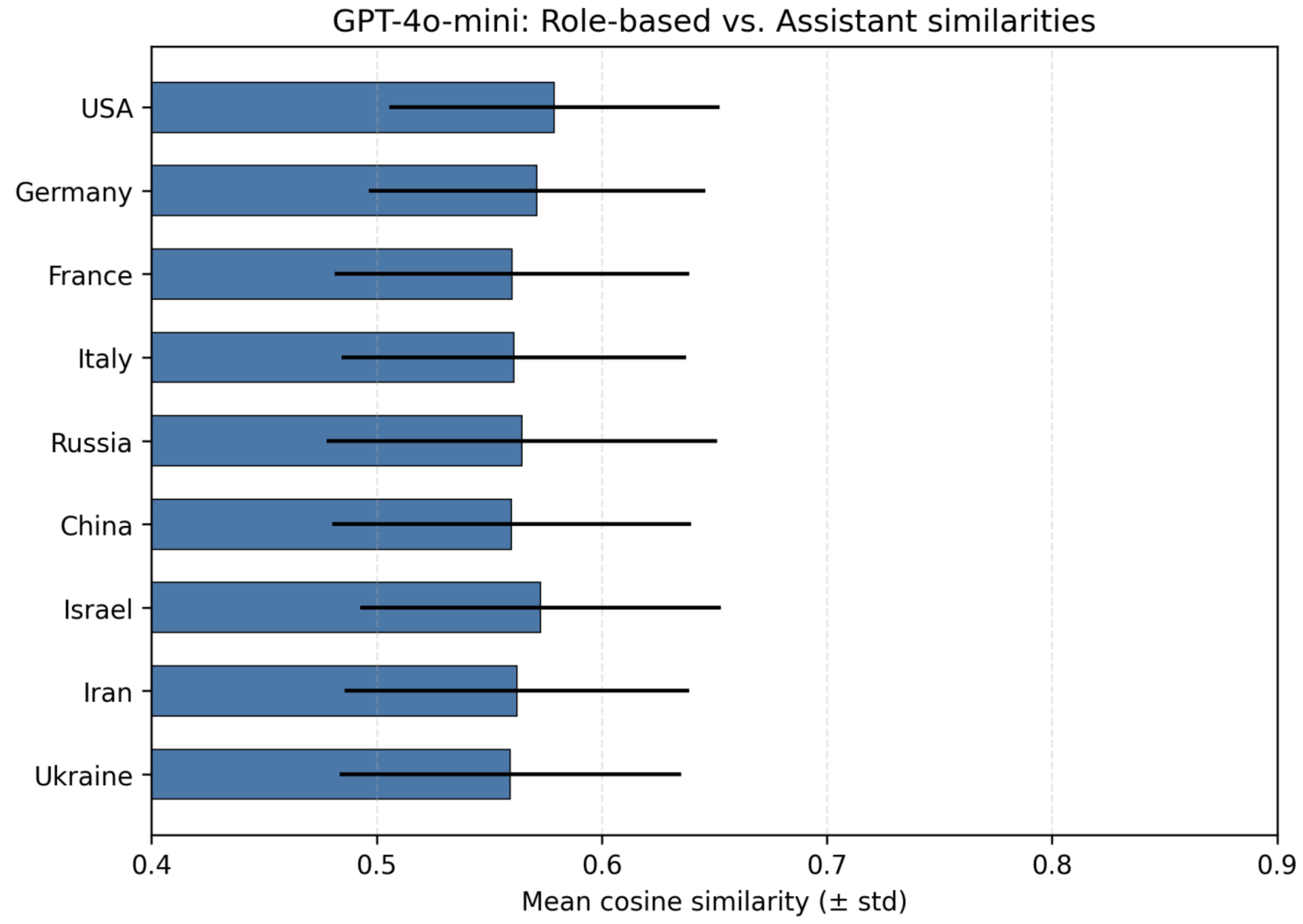
- LLaMA shows higher agreement with arguments.
- GPT agreement is more moderate and topic dependent.
- Native and Assistant conditions fluctuate more, suggesting stronger influence of topic and context.
- Role-based responses are the most consistent, showing less variation in agreement ratios.

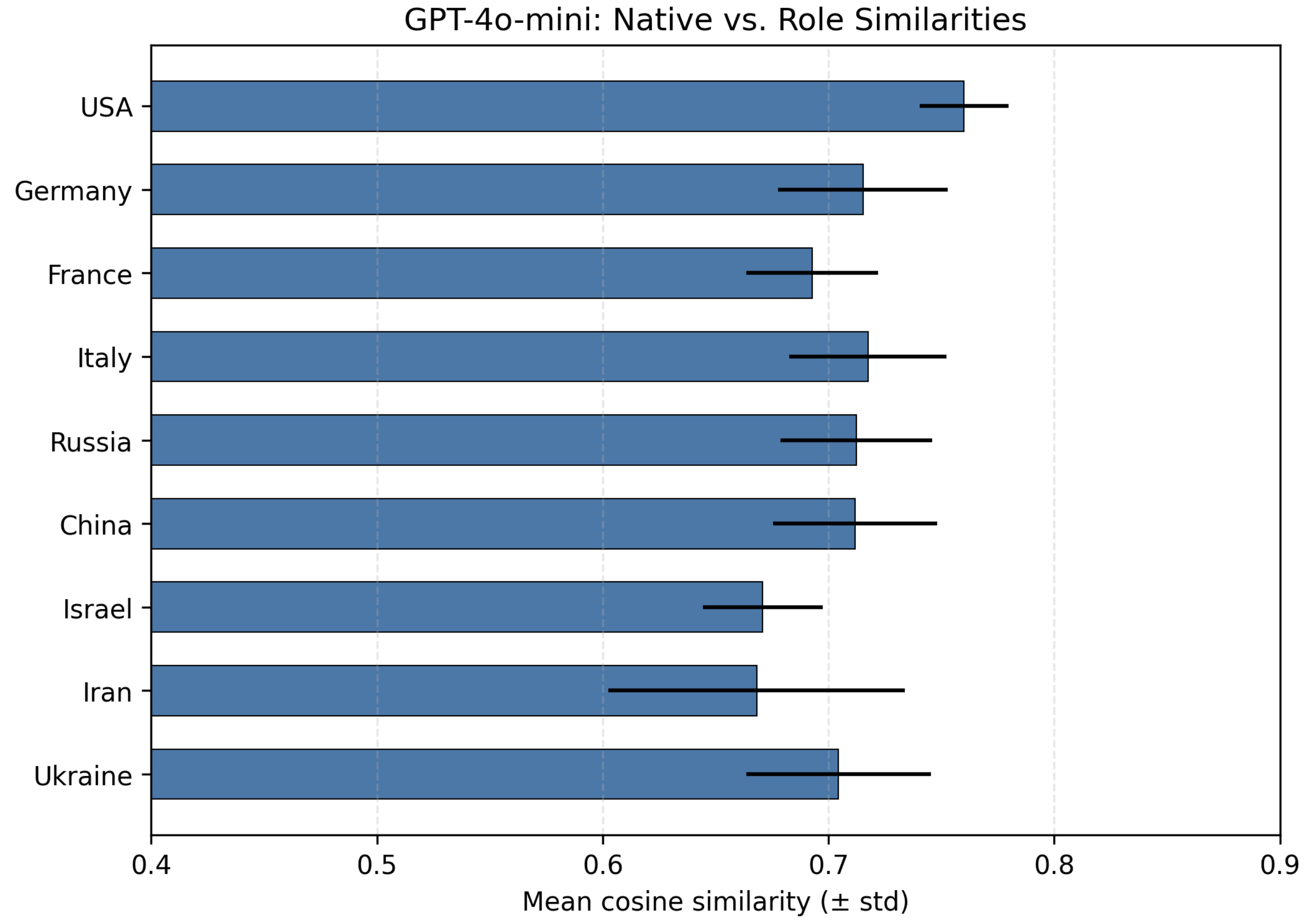


Results – Embedding similarities - GPT

- USA → highest
- Germany, France, Italy, Russia → mid–high
- China, Israel, Iran, Ukraine → lower, more variable
- Role prompts reshape reasoning more strongly than language, showing higher context sensitivity.

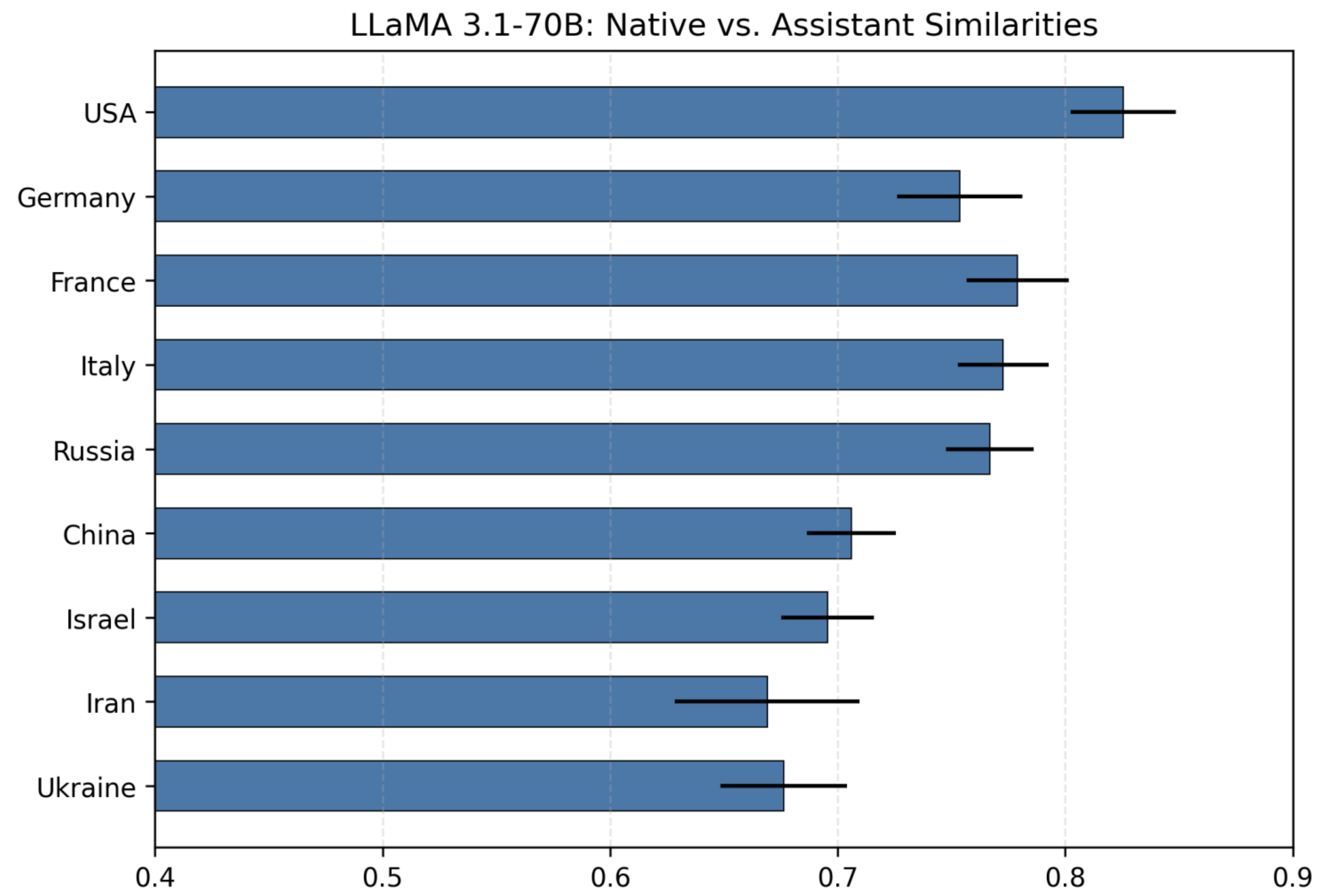


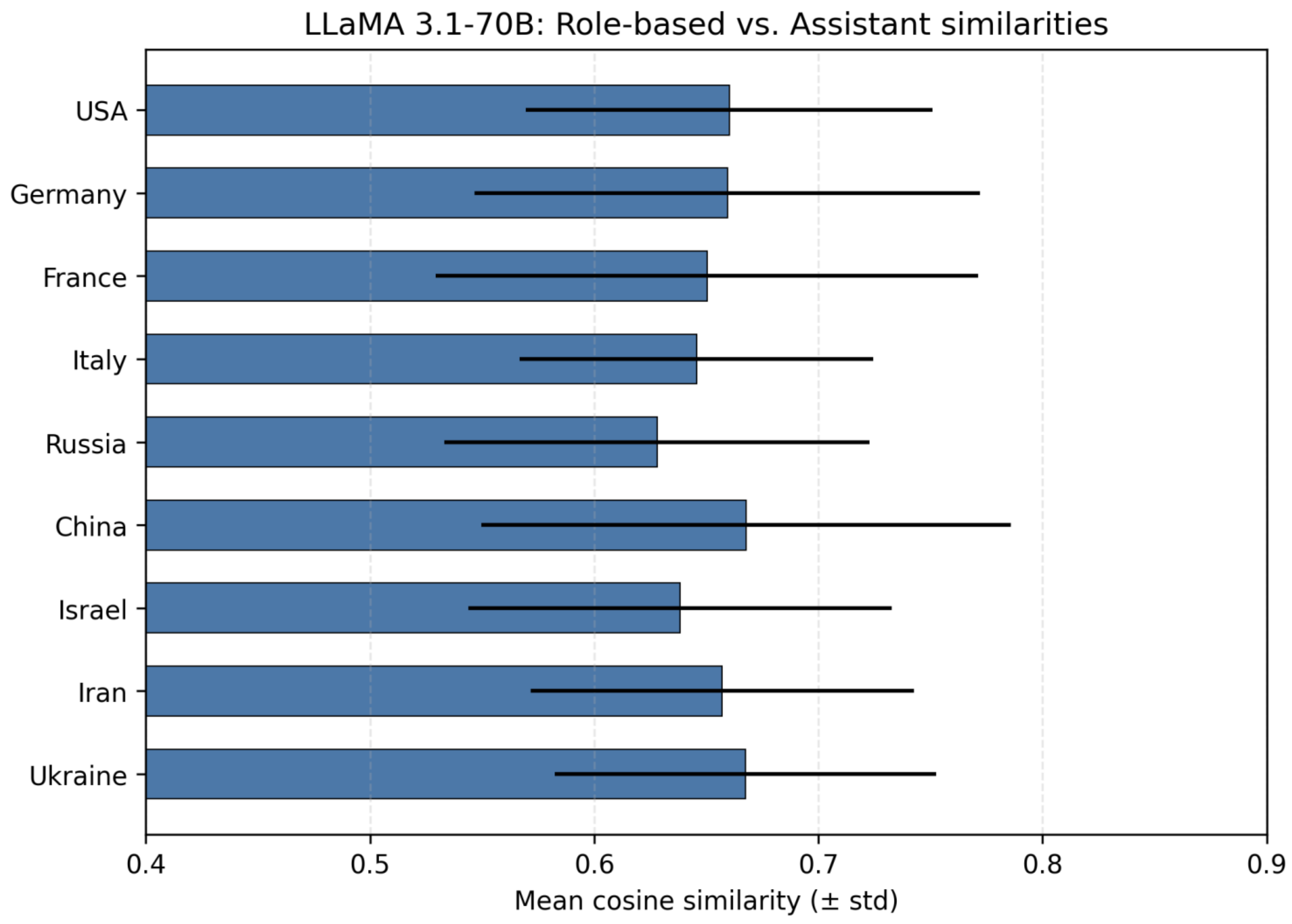


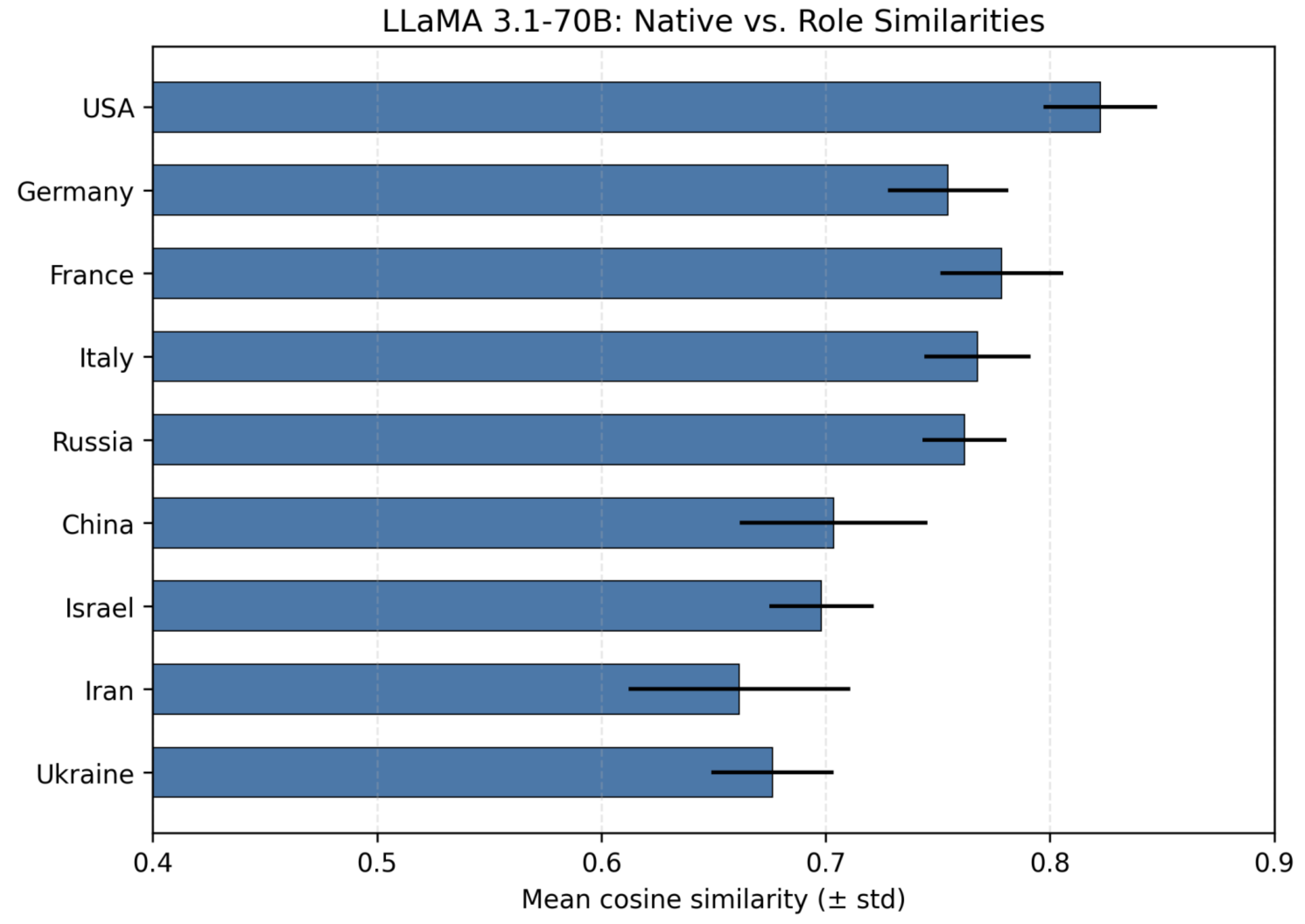


Results – Embedding similarities LLaMA

- USA → highest and most consistent
- China, Israel, Iran, Ukraine → lower, more variable
- France, Italy, Russia, Germany → medium and balanced
- Lower RvA → stronger role or persona effect







Conclusion

- **Are LLMs biased?**

  ✅ Yes — both models show ideological bias, but its strength depends on topic and framing
  multi-party system, *atheism* remained stable, *economic sanctions*, *voting*, *libertarianism* changed significantly across contexts.

- **Do they show ideological differences across roles and languages?**

  ✅ Yes — persona cues reshape reasoning far more than language changes.

- **Do different model families behave differently?**

  ✅ Yes — GPT-4o-mini is steadier and more consistent
  LLaMA 3.1-70B is more variable and sensitive

Limitations & future work

- Test intra-model consistency through repeated prompting.
- Investigate potential effects of model access location on responses.
- Compare with external reference sources (e.g., Wikipedia)

Thank You for Listening!
Happy to Take your Questions.

UNIVERSITÀ
DEGLI STUDI
DI MILANO

Agreement Ratio

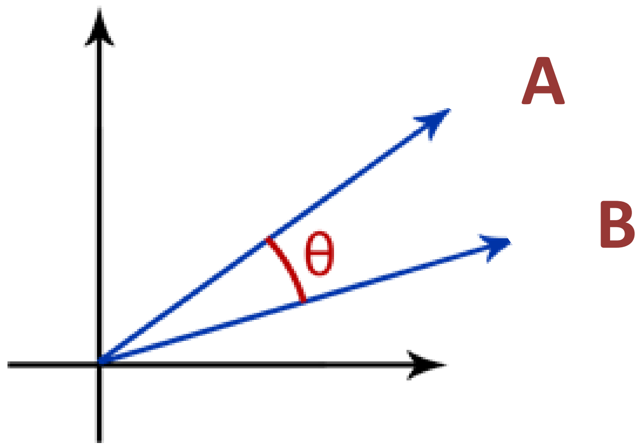
Model stance	Argument stance	Model $\rightarrow$ Conclusion
Agree	In favor of	Agree
Disagree	In favor of	Disagree
Agree	Against	Disagree
Disagree	Against	Agree

Embedding Similarities

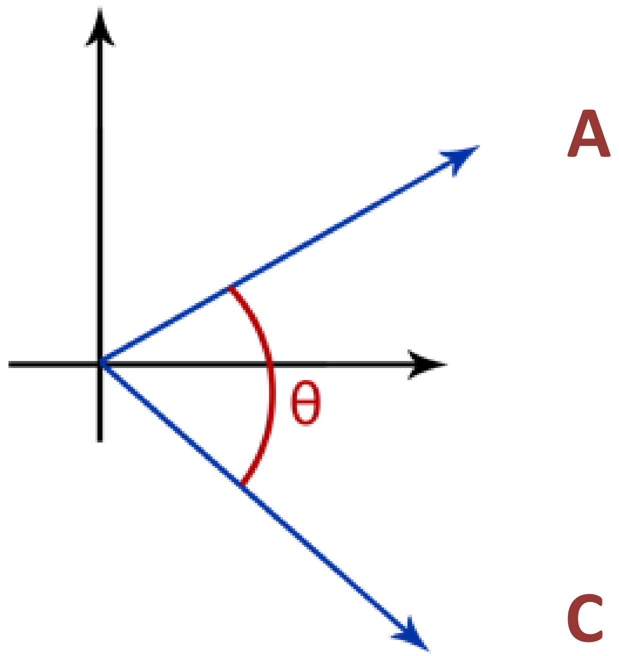
A = I support libertarianism because individual freedom should come before government control.

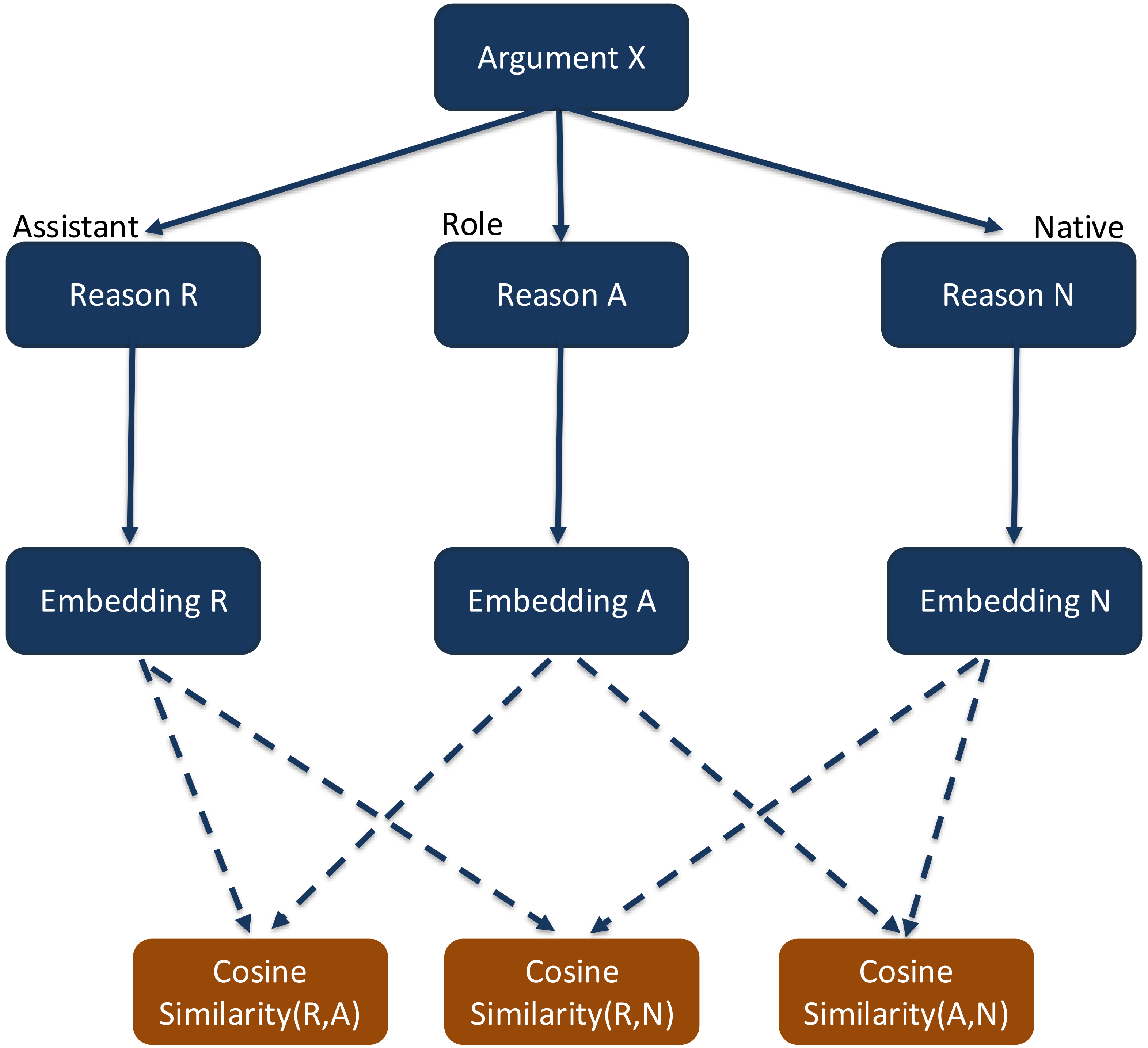
B = People should be free to make their own choices without unnecessary state interference.

C = I disagree with libertarianism since too little regulation can harm social equality and public welfare.



$$sim(A,B) = \cos(\theta) = \frac{A \cdot B}{\|A\|\|B\|}$$



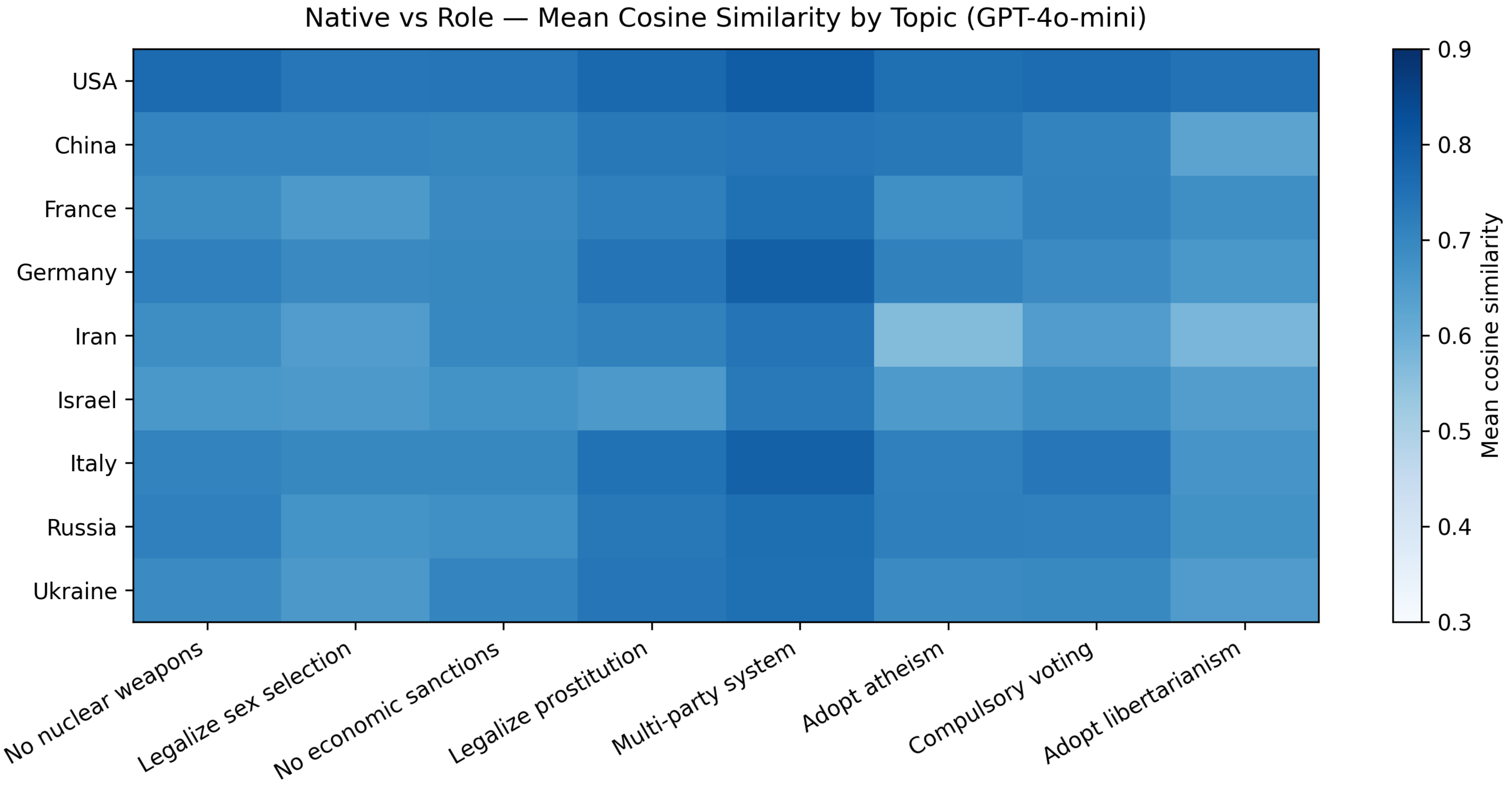


Standard deviations

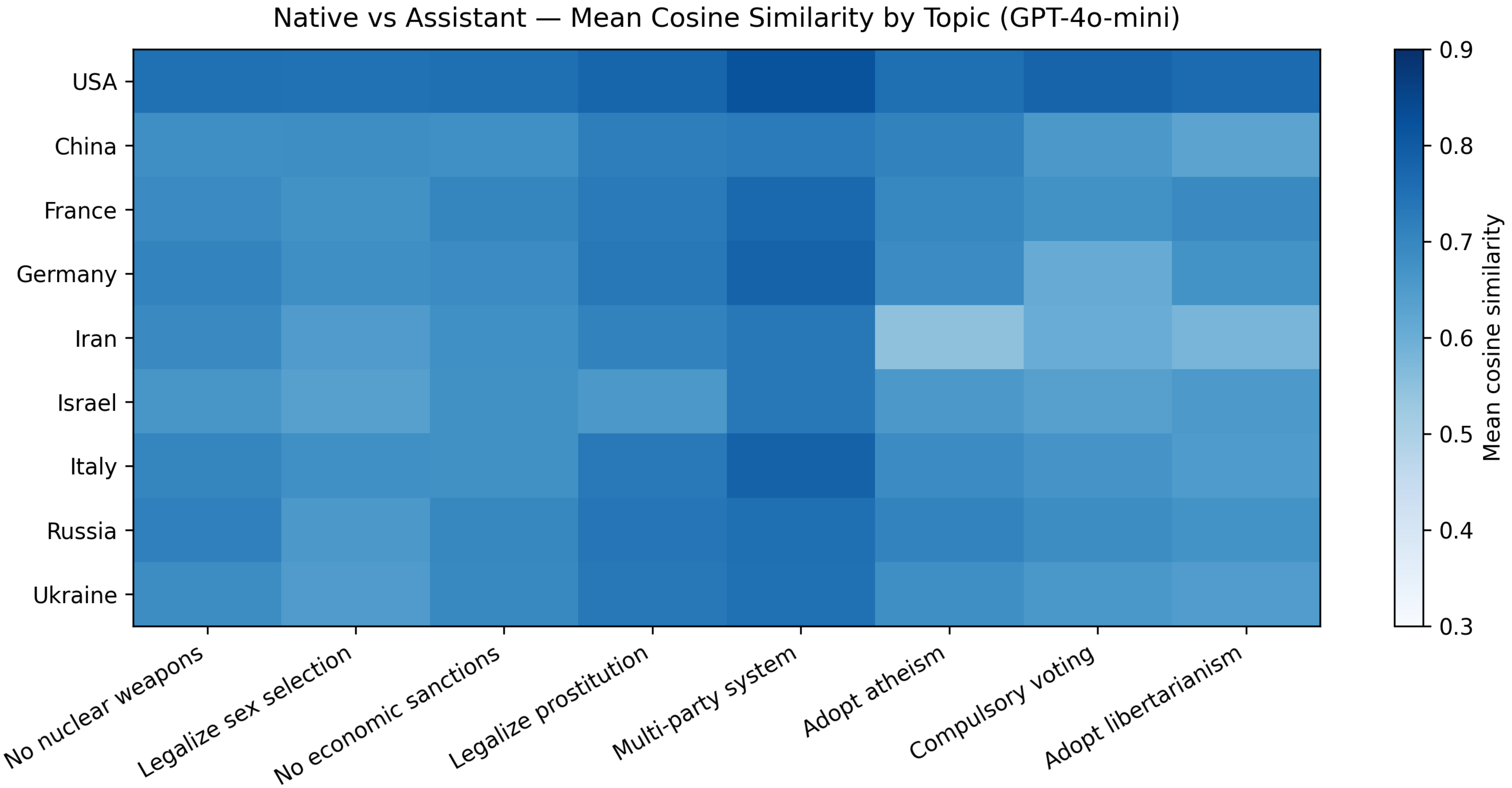
- σ_c: The **standard deviation** of the cosine similarities for country c. This measures how spread out the similarity scores are across different topics within that country.
- N_c: The **number of topics** for country c. This is the count of data points being analyzed.
- $\text{sim}_{c,t}$: The **mean cosine similarity of topic** t in country c. This is the individual data point whose variation is being measured.
- $\overline{\text{sim}_c}$: The **average similarity across all topics** for country c. This is the mean (average) of all the $\text{sim}_{c,t}$ values.

$$\sigma_c = \sqrt{\frac{1}{N_c - 1} \sum_{t=1}^{N_c} (sim_{c,t} - \overline{sim_c})^2}$$

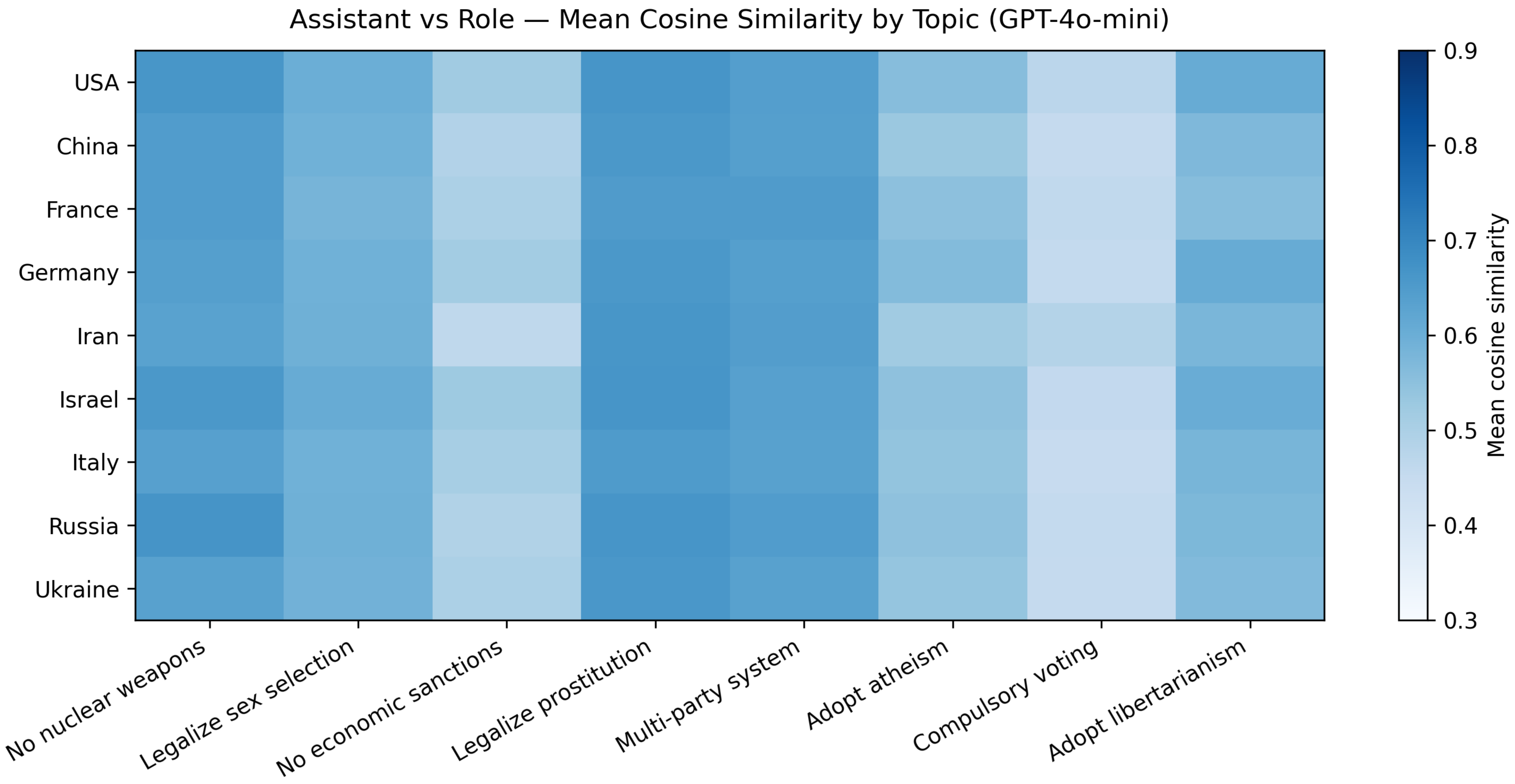
Similarities – GPT - NvR



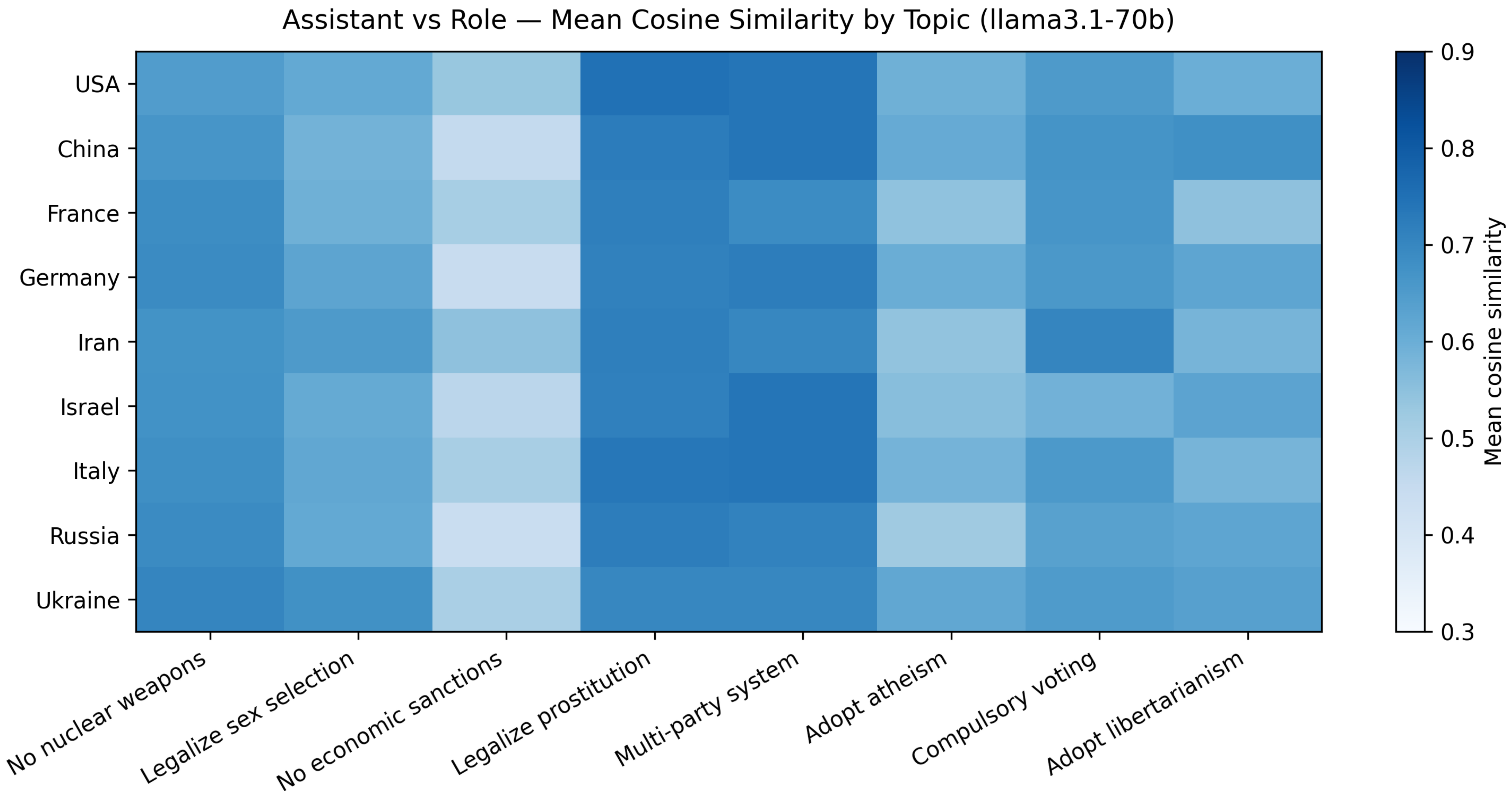
Similarities – GPT - NvA



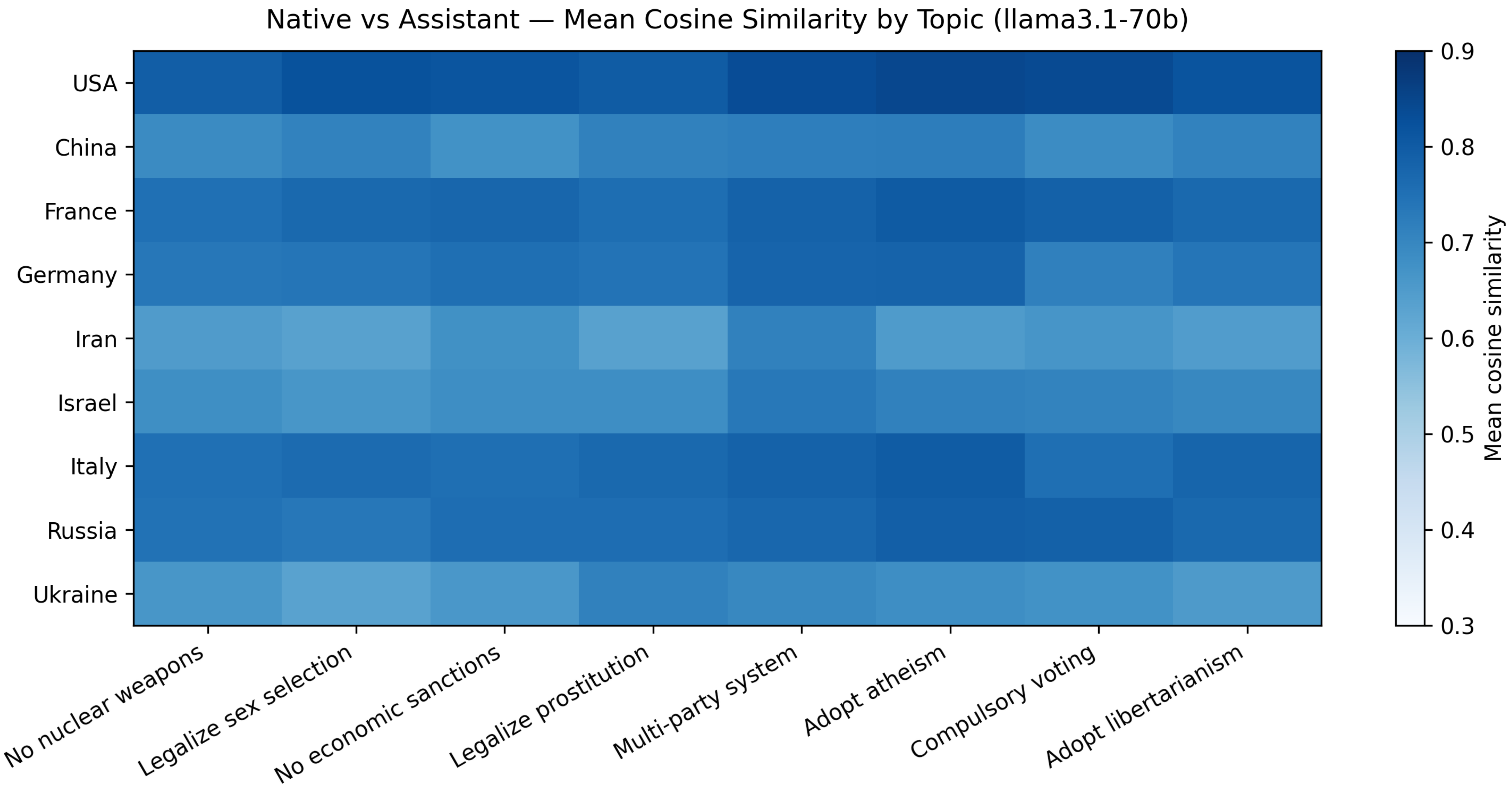
Similarities – GPT - RvA



Similarities – LLaMA - RvA



Similarities – LLaMA - NvA



Similarities – LLaMA - NvR